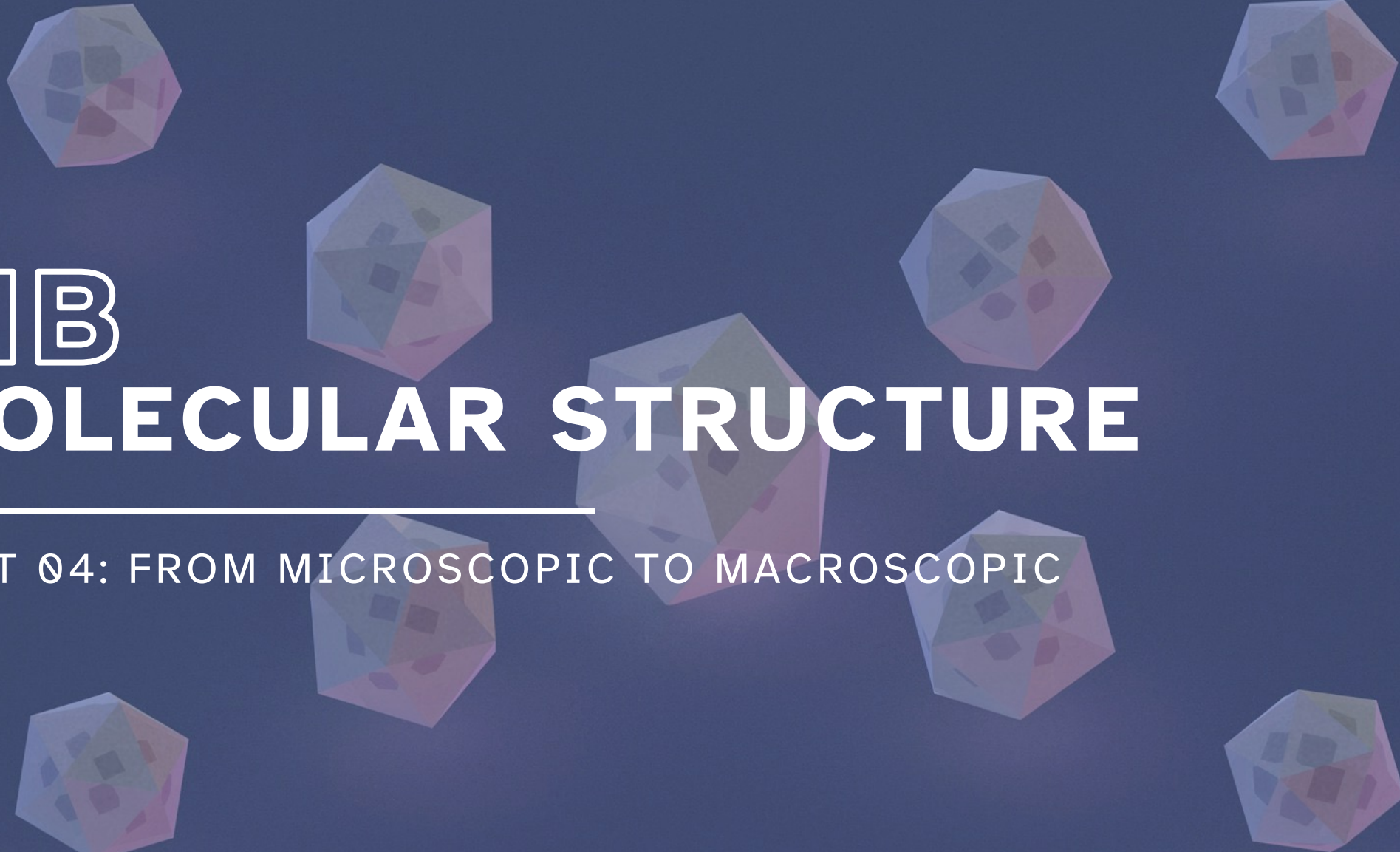


The background features several translucent, multi-colored geometric shapes, primarily cubes and octahedrons, scattered across the dark blue gradient. These shapes have internal patterns and are semi-transparent, allowing the background to show through them.

11B MOLECULAR STRUCTURE

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

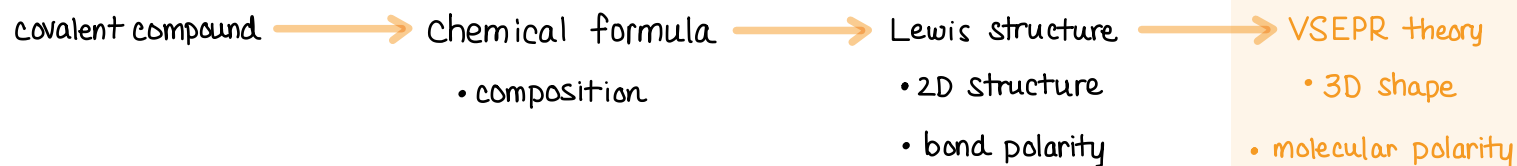
CHAPTER 11

1. Use VSEPR theory and Lewis structures to predict molecular shapes.
2. Identify polar and nonpolar molecules using electronegativity values and molecular shape.
3. Explain valence bond theory.
4. Describe and recognize sp , sp^2 , and sp^3 hybridization based on the Lewis structure of a molecule.
5. Define, describe, and identify sigma (σ -) and pi (π -) bonds in molecules.

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



Polar and Nonpolar Molecules

11.2

Molecular polarity

- Molecules can be comprised of many polar and/or nonpolar covalent bonds.
- The polarity of the bonds, or bond dipoles, in a molecule can also give rise to a net or overall molecular dipole. We call this molecular polarity.
- The polarity of a molecule can be predicted by analyzing:
 - 1) Relative electronegativities
 - 2) Bond polarities or bond dipoles
 - 3) Molecular shapes and bond angles

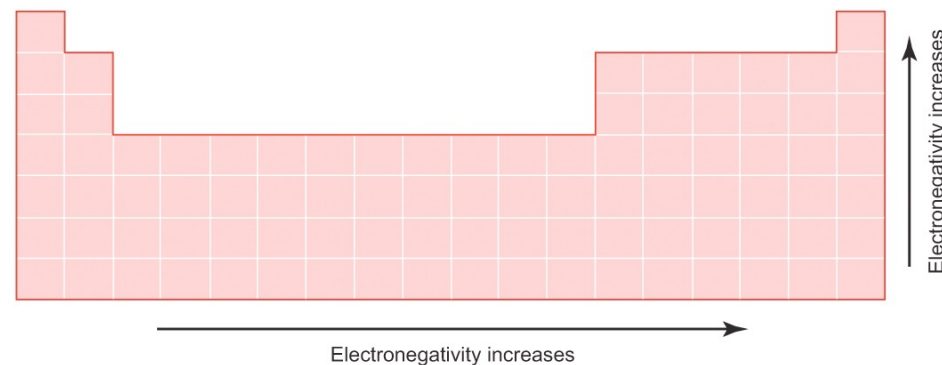


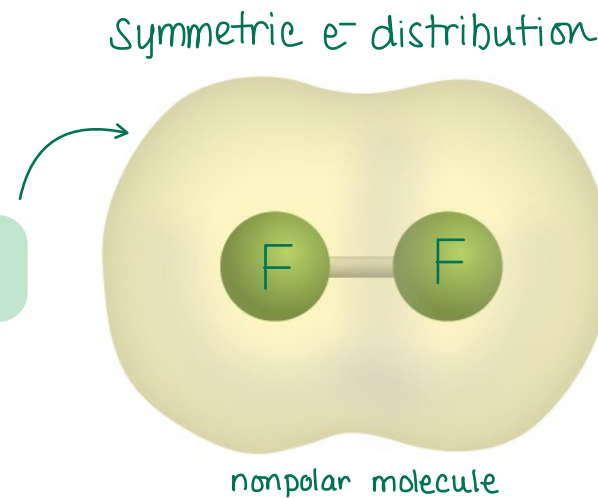
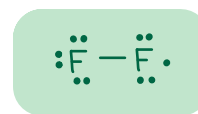
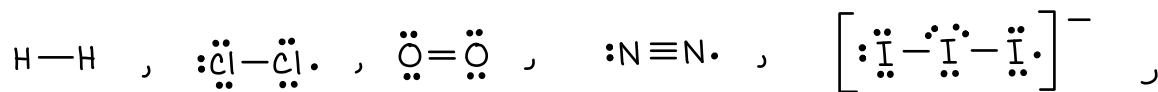
FIGURE 11.15 Electronegativity Trend

Polar and Nonpolar Molecules

11.2

Case 1: Molecules with ONLY nonpolar bonds

- These will *always* be *nonpolar molecules*.
- Bonds between the *same* elements (e.g., C–C, H–H, etc.) are nonpolar bonds.
 - No difference in electronegativity between the two atoms: $\Delta\text{EN} = 0$
- There are *no bond dipoles* and, consequently, *no net molecular dipole*.
- *Examples*) H_2 , F_2 , O_2 , N_2 , and I_3^- are nonpolar molecules

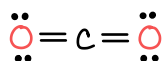


Polar and Nonpolar Molecules

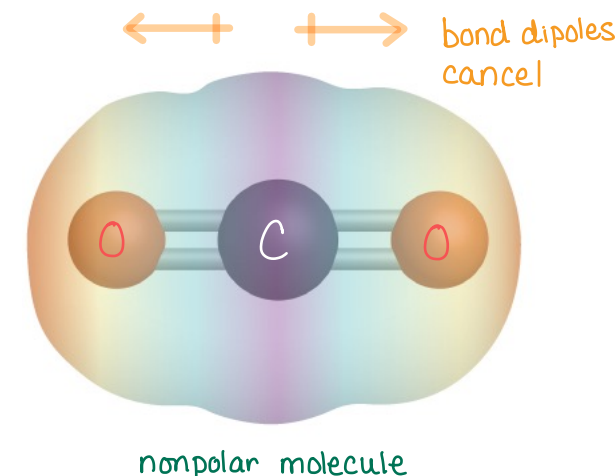
11.2

Case 2: Molecules with polar bonds that are SYMMETRIC

- Bonds between *different* elements (e.g., C–O, O–H, etc.) are polar bonds.
 - There is a difference in electronegativity between the two atoms: $\Delta EN \neq 0$
- Bond dipoles exist, but they cancel out (i.e., the vectors sum to zero) because of the *symmetric shape* of the molecule.
- There is *no net molecular dipole*, so these are *nonpolar molecules*.
- Examples) CH₄, BF₃, and CO₂ are nonpolar molecules.



Symmetric e⁻ distribution

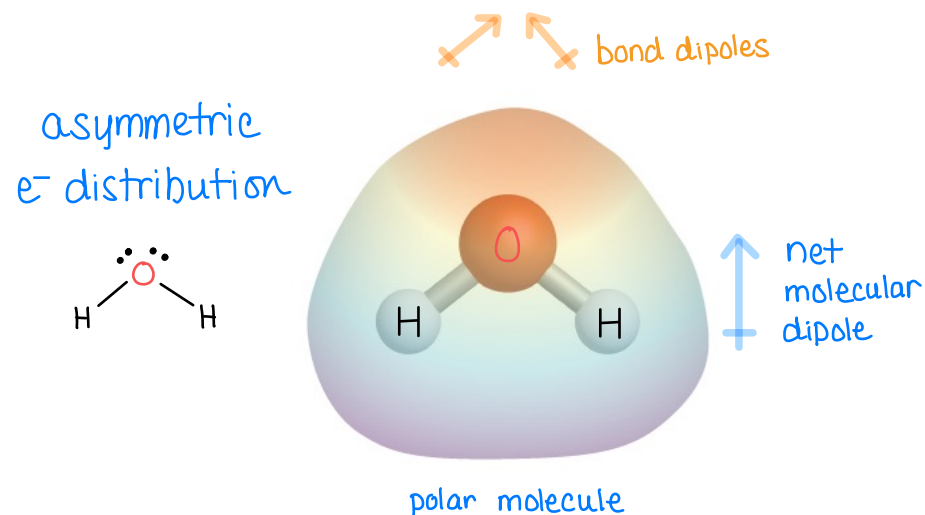
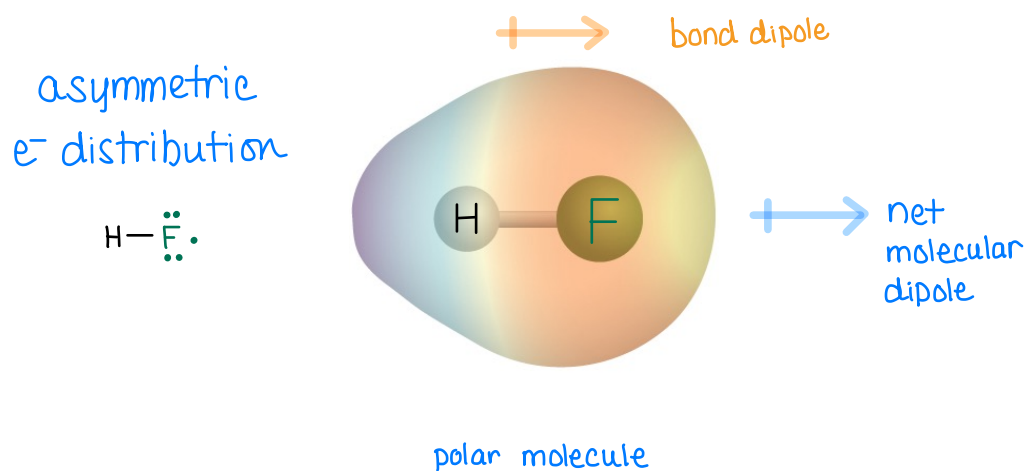


Polar and Nonpolar Molecules

11.2

Case 3: Molecules with polar bonds that are ASYMMETRIC

- The bond dipoles do NOT cancel (i.e., vectors do not sum to zero) because of the *asymmetric shape* of the molecule.
- The bond dipoles add/combine together to form a *net molecular dipole*, so these are *polar molecules*.
- *Examples*) HF, H₂O, and NF₃ are polar molecules.



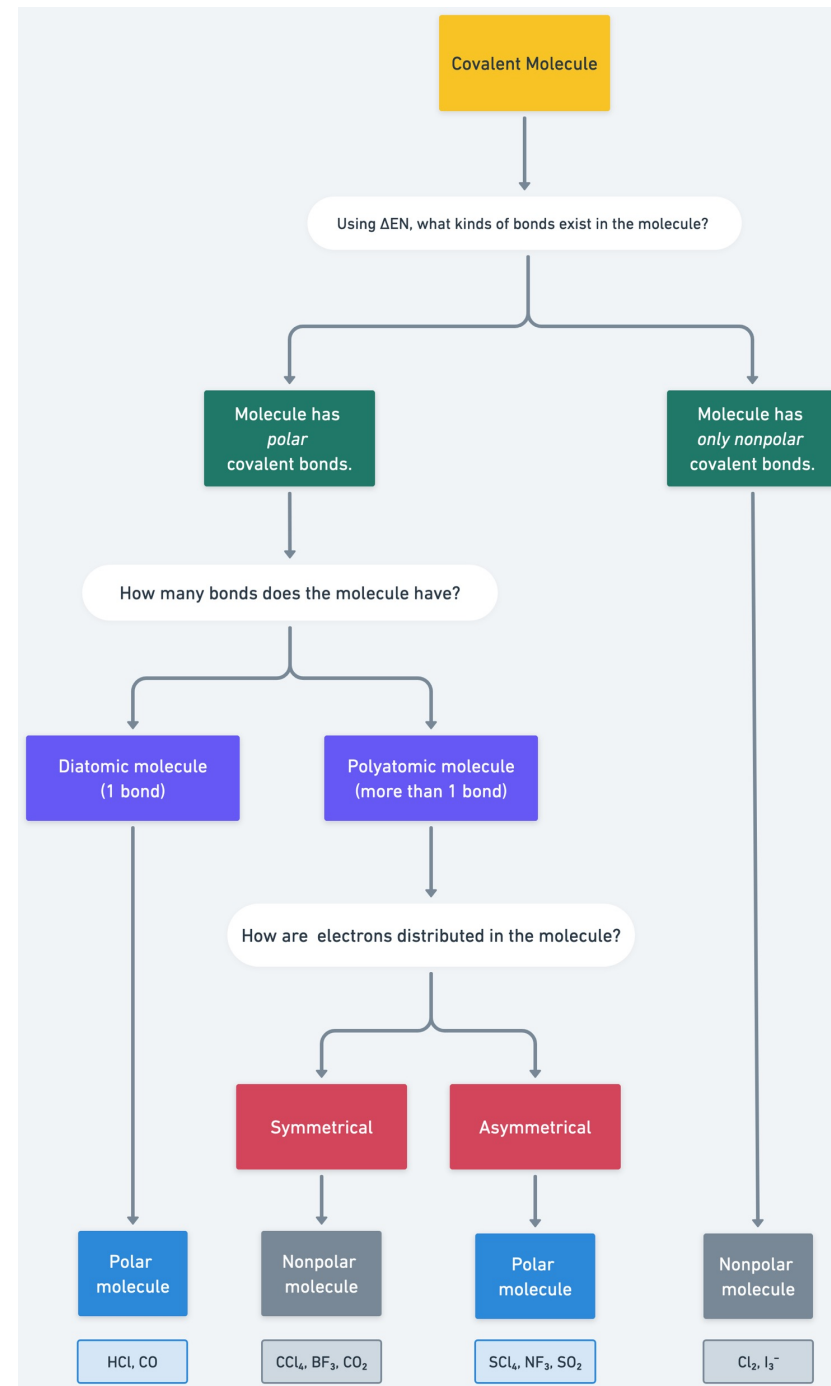
Polar and Nonpolar Molecules

11.2

Molecular polarity depends on the shape of the molecules and the bond angles.

There are two ways to determine molecular polarity:

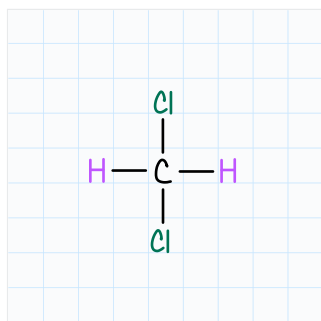
- 1) Mathematically, add the bond dipole vectors to see if a **net molecular dipole** exists.
 - 2) Visually, inspect the symmetry of the electron distribution throughout the molecule. Use flowchart to the right.
- Symmetric molecules tend to be nonpolar.
 - Asymmetric molecules tend to be polar.



EXAMPLE PROBLEM

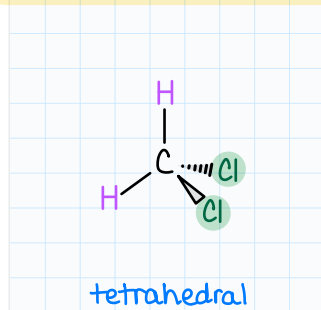
Learning Objective(s): 11.2

Is CH_2Cl_2 a polar or nonpolar molecule?



cannot rely on just Lewis structure!
this looks "symmetric" b/c it's not 3D!

all angles are 109.5° , so not planar!

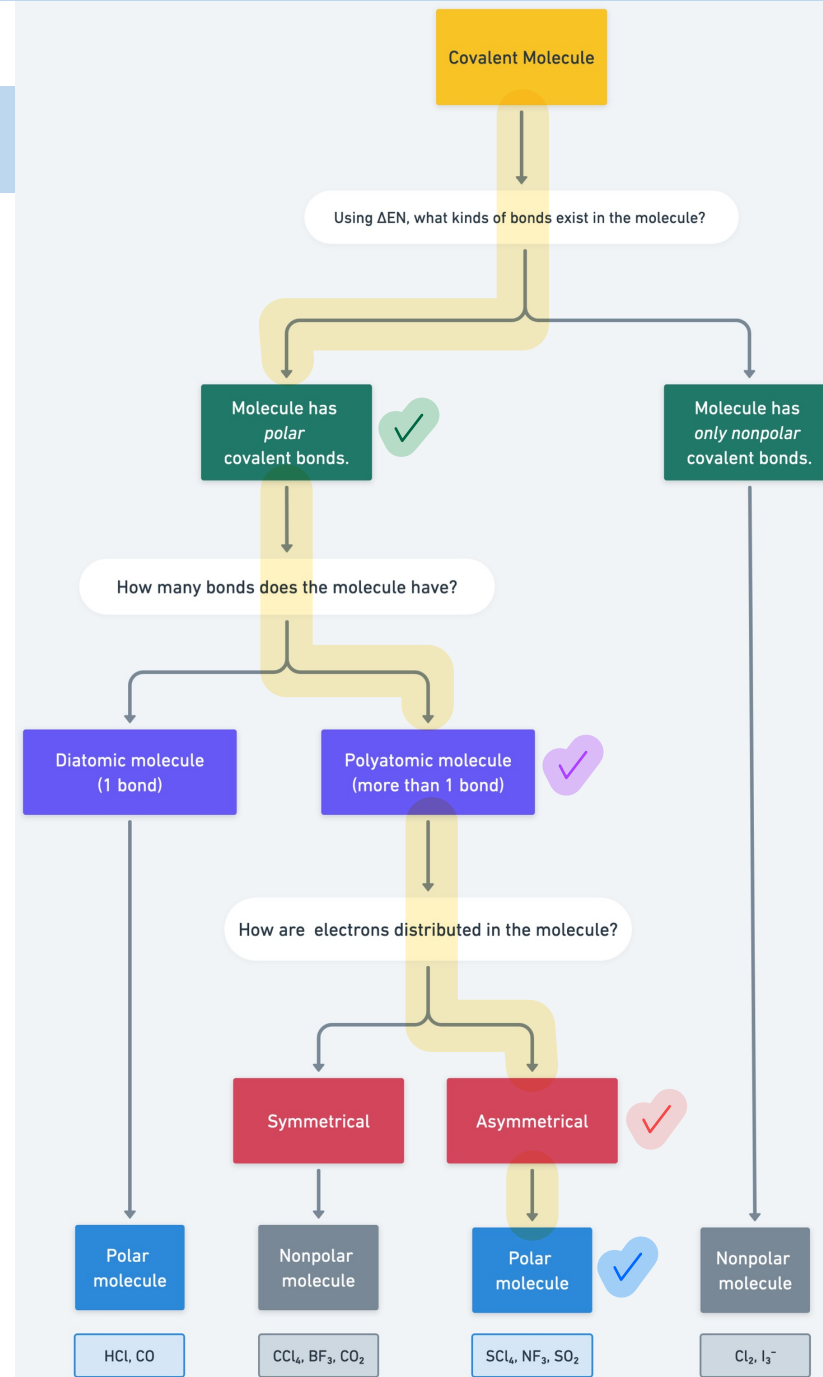


but it's actually asymmetric:

- the Cl side will have more of e^- distribution
- C-Cl bond dipoles will never cancel
- C-H bond dipoles will never cancel

polar molecule

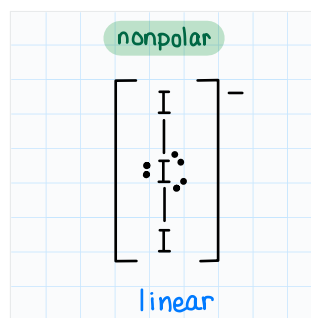
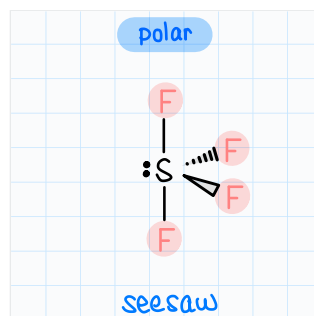
note: lone pairs on Cl omitted for clarity



EXAMPLE PROBLEM

Learning Objective(s): 11.2

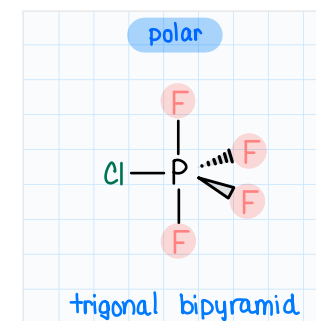
Predict whether each of the following molecules is polar or nonpolar.



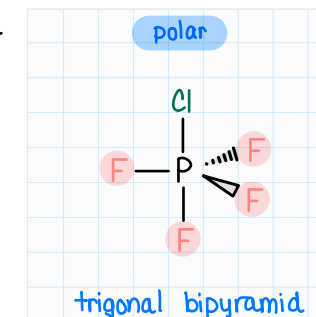
* only nonpolar bonds!



structure 1



structure 2



◄ bond coming out of plane of paper

..... bond going into plane of paper

for clarity, lone pairs not shown on outer/terminal atoms.

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

When two atoms approach each other, their electrons and nuclei interact.

- If the *energy* of system is *lowered* due to the atoms interacting, then we say that a *covalent bond forms* between the two atoms.

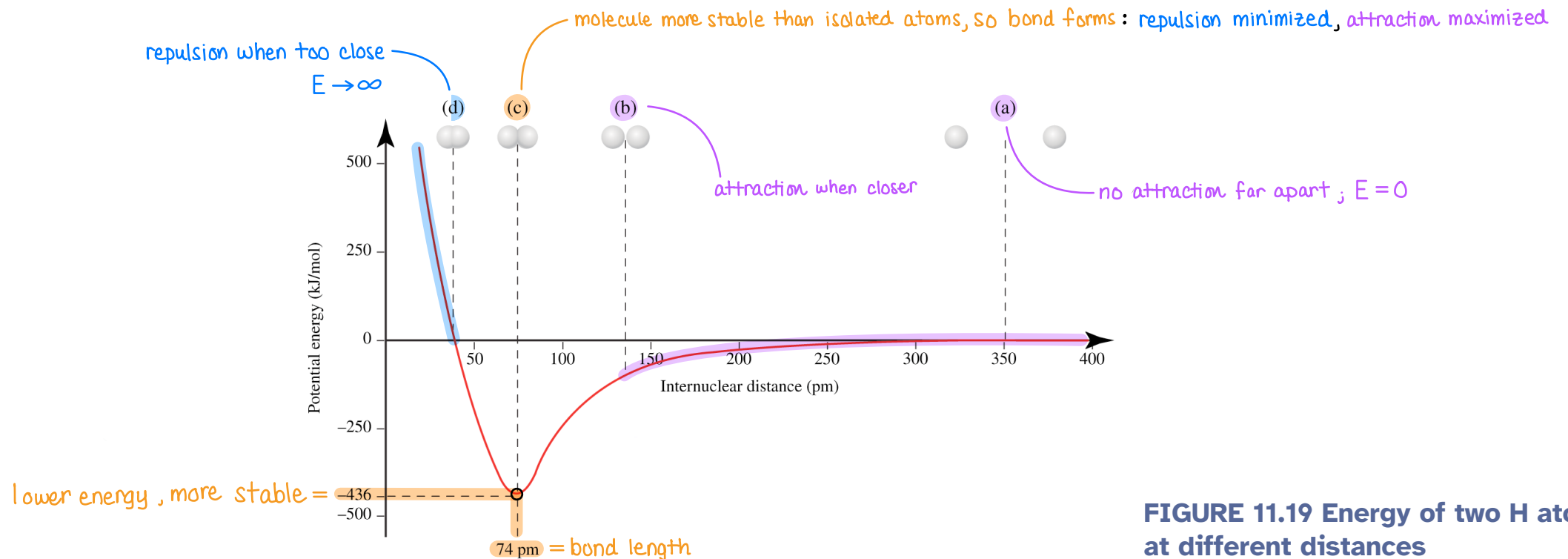


FIGURE 11.19 Energy of two H atoms at different distances

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

Valence Bond (VB) Theory

- Describes *how* covalent bonding takes place.
- This involves the overlap of orbitals on adjacent atoms.

The *simplest* form of VB theory uses *only* the standard atomic orbitals (AOs) to describe the bonding in many **diatomic molecules** very well.

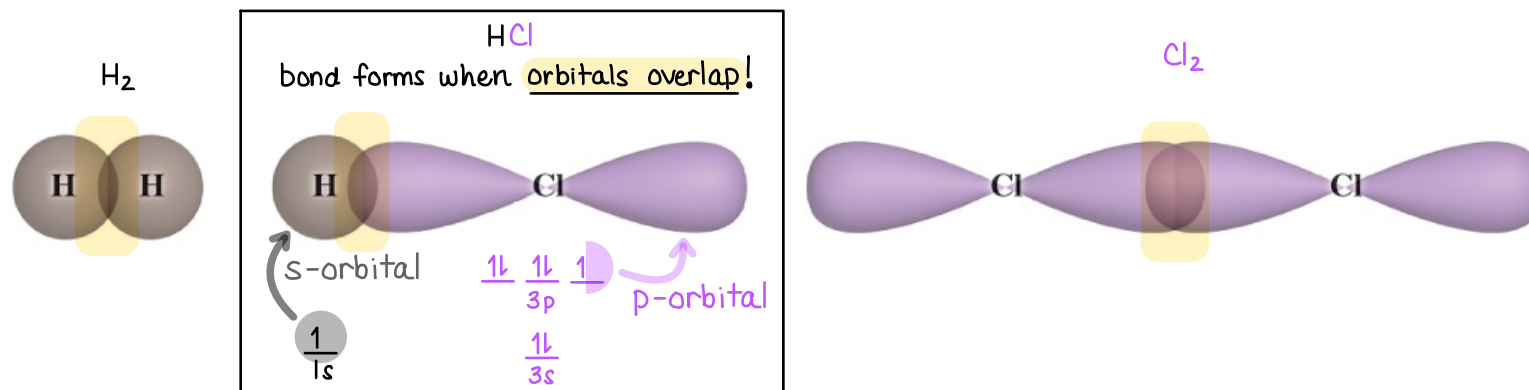
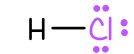


FIGURE 11.25 (similar)

Lewis structure



VSEPR theory

linear electron geometry

linear molecular shape, 180°

VB theory

- half-filled $1s$ on H atom overlaps with half-filled $3p$ on Cl atom
- predicts: linear shape

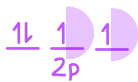
Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

Unfortunately, simple VB theory (using *only* atomic orbitals) fails very quickly.

For example, simple VB theory *erroneously* predicts a 90° bond angle in H_2O .

- An **O atom** with two half-filled **$2p$ -orbitals** overlaps with a half-filled **$1s$ -orbital** on each H atom.



Lewis structure

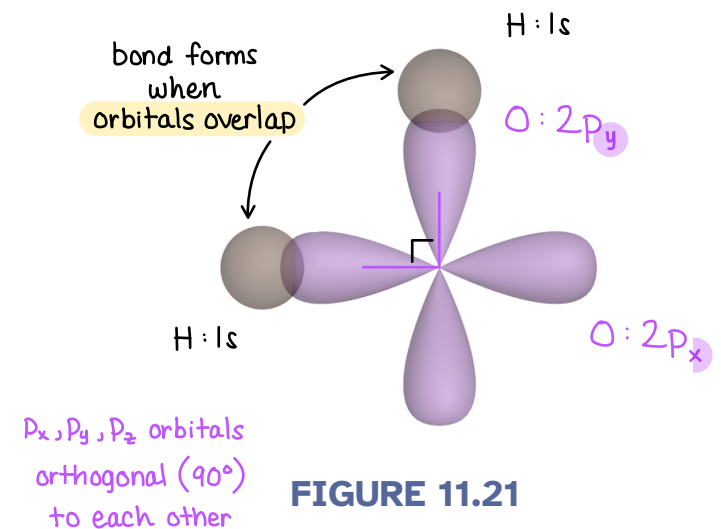
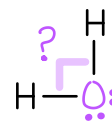


VSEPR theory

tetrahedral electron geometry
bent molecular shape, $\sim 105^\circ$

VB theory

- predicts: 90° bond angle?
- how to fix theory to explain
observed/real $\sim 105^\circ$?



Valence Bond Theory: Hybrid Orbitals and Bonding

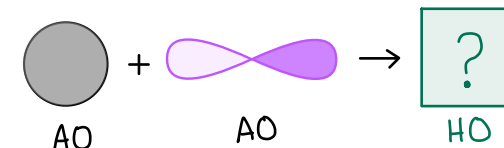
11.3

Hybrid orbitals (HOs)

- Used to explain covalent bonding for *some VSEPR molecular shapes*.
- **HOs** constructed by linear combinations of atomic orbitals (AOs).

Recall: All orbitals are just mathematical wavefunctions.

- AOs involved in hybridization: *only* **s**- and **p**-orbitals.
- Shapes and energies of HOs are *different* from their constituent AOs.



When to use hybridization? When the total number of domains on the central atom is 2, 3, or 4; *i.e.*, when electron geometry is linear, trigonal planar, or tetrahedral.



When NOT to use hybridization? When the total number of domains on the central atom is 5 or 6; *i.e.*, when electron geometry is trigonal bipyramidal or octahedral.

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

Central atoms are most likely to hybridize.

- Number of electron domains determines the type of hybridization.
- Each electron domain represents ONE hybrid orbital (HO).
- Number HOs = number AOs mixed

Electron Geometry	# of Domains	Hybridization Type	# and Types of AOs Mixed	# and Types of HOs Made	# of Leftover Unhybridized <i>p</i> -orbitals
tetrahedral	four	sp^3	$s + p + p + p$ (one <i>s</i> - and three <i>p</i> -orbitals)	four sp^3 HOs	none
trigonal planar	three	sp^2	$s + p + p$ (one <i>s</i> - and two <i>p</i> -orbitals)	three sp^2 HOs	one <i>p</i> -orbital
linear	two	sp	$s + p$ (one <i>s</i> - and one <i>p</i> -orbital)	two sp HOs	two <i>p</i> -orbitals

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

sp^3 hybridization

- If a molecule has four electron domains, then four hybrid orbitals are made from four atomic orbitals of the central atom.
- The four sp^3 hybrid orbitals overlap at their center (nodes) to form the characteristic tetrahedral electron geometry.

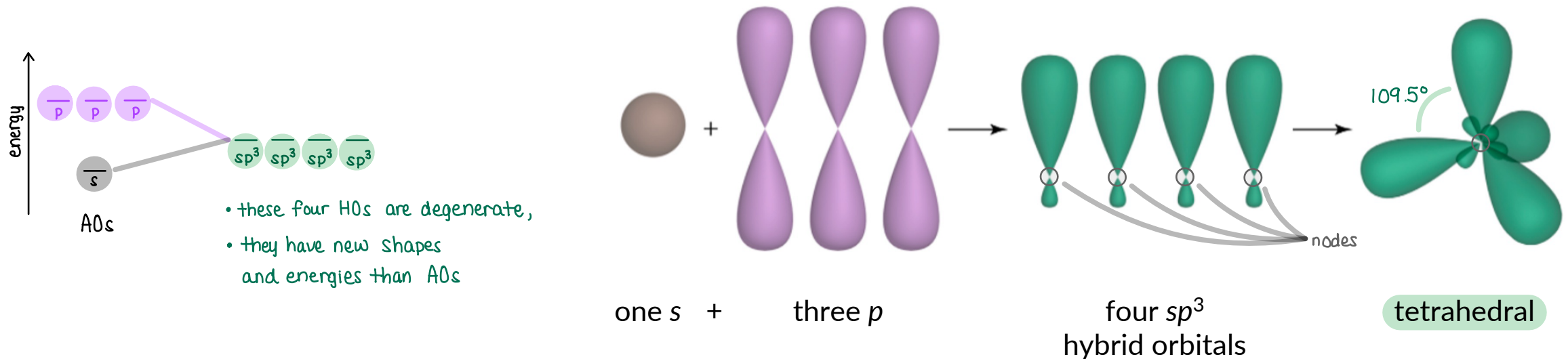


FIGURE 11.23 (similar)

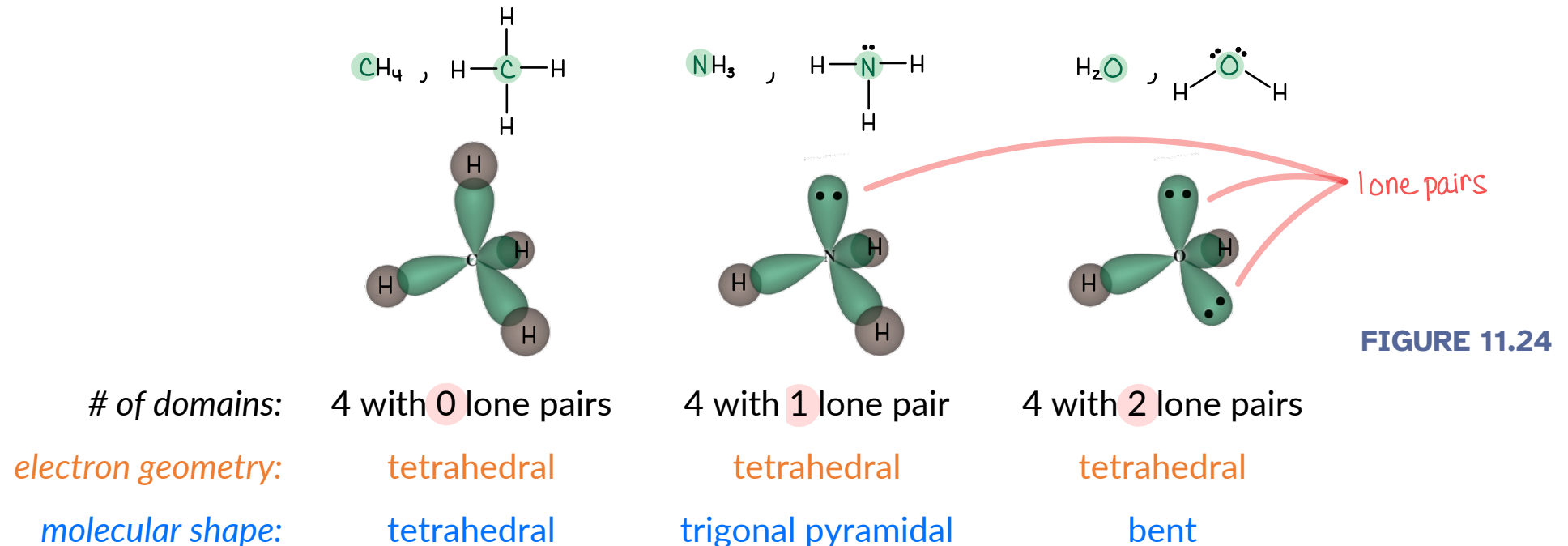
Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

sp^3 hybridization

smaller lobe not shown for clarity

- The larger lobe of each sp^3 hybrid orbital can overlap with another atomic or hybrid orbitals to form bonds.
- Or lone pairs of electrons can occupy the sp^3 hybrid orbitals.



Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

sp^2 hybridization

- If a molecule has three electron domains, then three hybrid orbitals are made from three atomic orbitals of the central atom.
- The three sp^2 hybrid orbitals form the trigonal planar electron geometry.

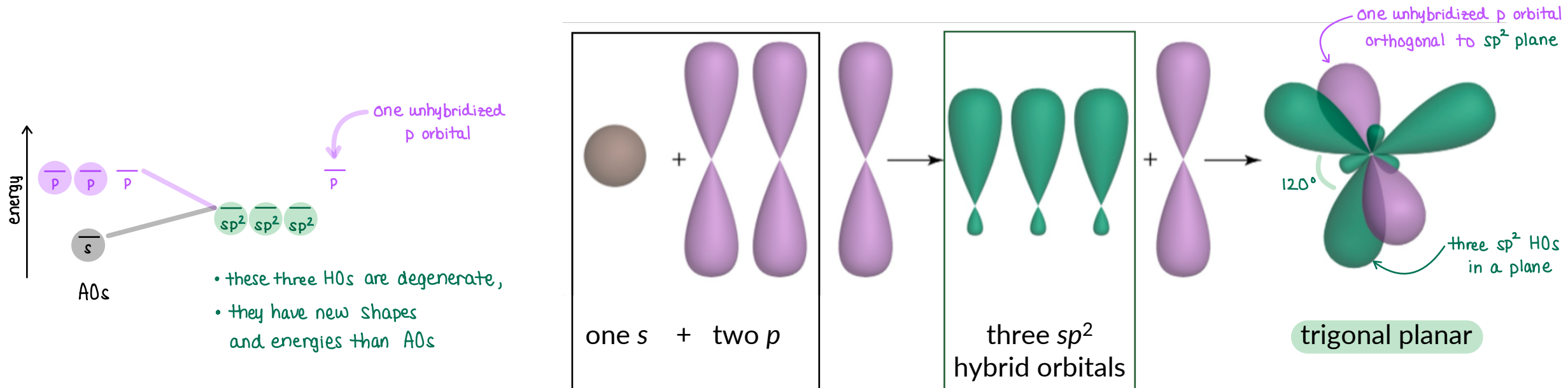


FIGURE 11.26 (similar)

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

sp hybridization

- If a molecule has two electron domains, then two hybrid orbitals are made from two atomic orbitals of the central atom.
- The two *sp* hybrid orbitals form the linear electron geometry.

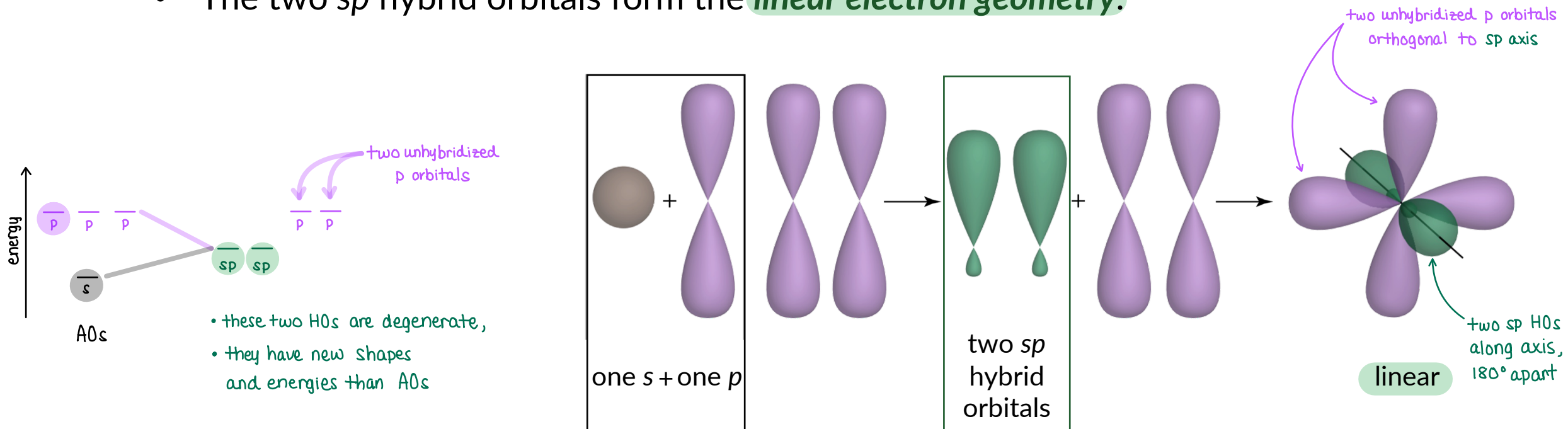


FIGURE 11.29 (similar)

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

Sigma (σ -) bonds

- Formed via constructive/additive overlap of orbitals.
- Direct or head-on overlap of orbitals along the internuclear axis.
- All bonds (single, double, or triple) have one σ -bond.

many ways to form σ -bonds

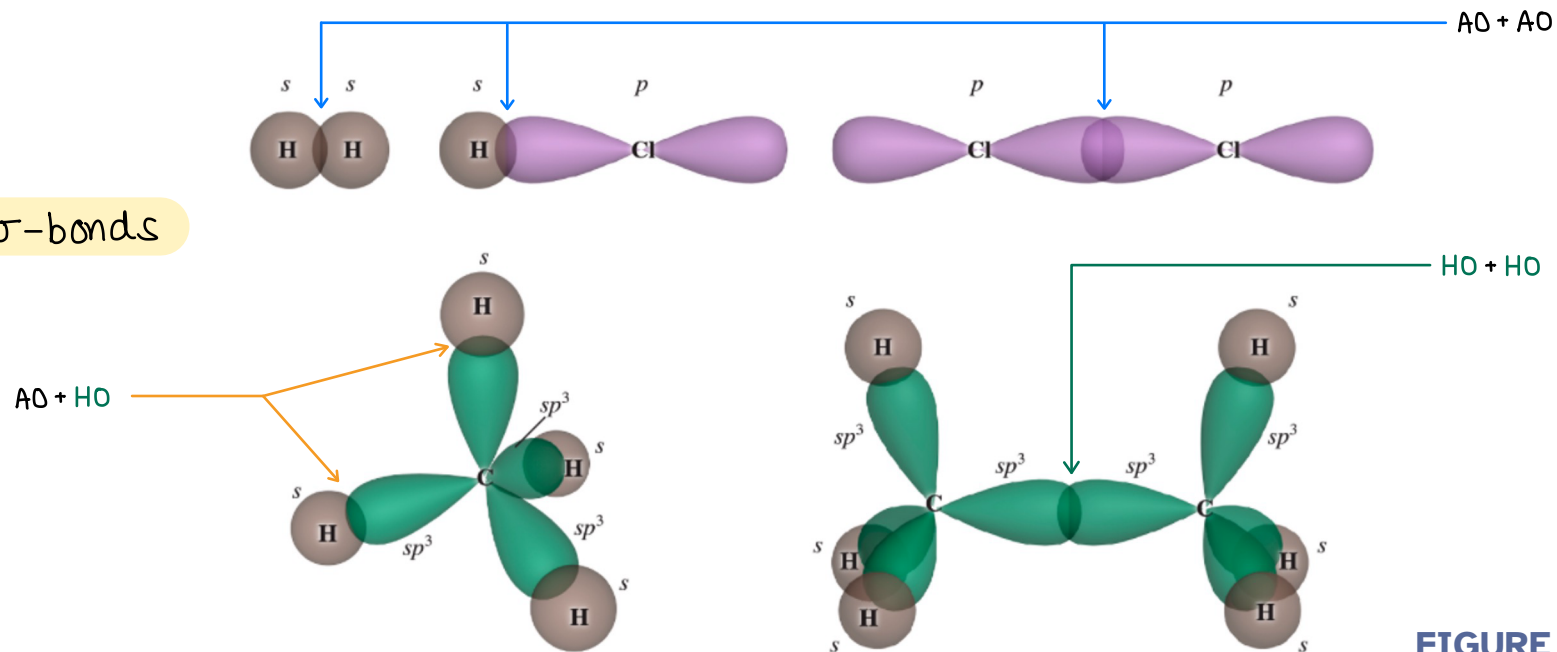


FIGURE 11.25 (similar)

Valence Bond Theory: Hybrid Orbitals and Bonding

11.3

Pi (π -) bonds

- Formed via constructive/additive overlap of unhybridized p -orbitals.
- Only sp^2 - and sp -hybridized atoms can exhibit π -bonding.
- Parallel or side-to-side overlap of p -orbitals, *above* and *below* internuclear axis.
- Double bond = one σ -bond + one π -bond
- Triple bond = one σ -bond + two π -bonds

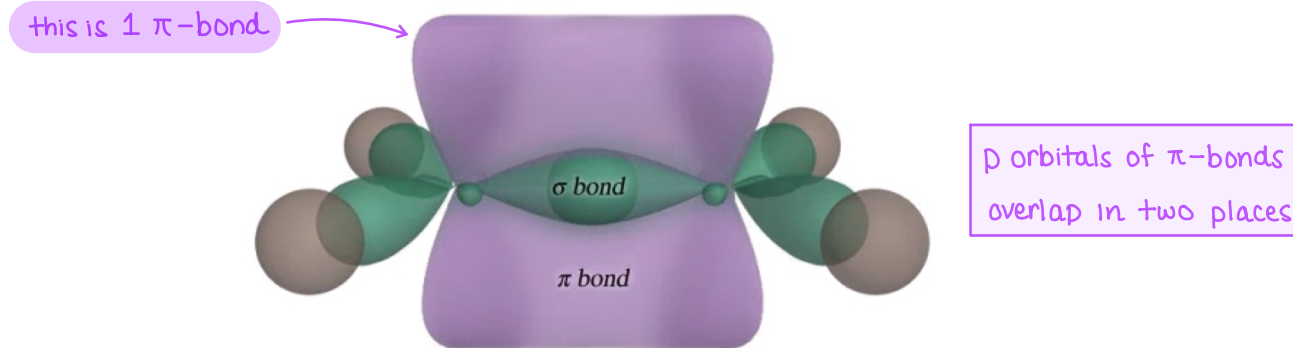


FIGURE 11.27 One π -bond in $\text{H}_2\text{C}=\text{CH}_2$

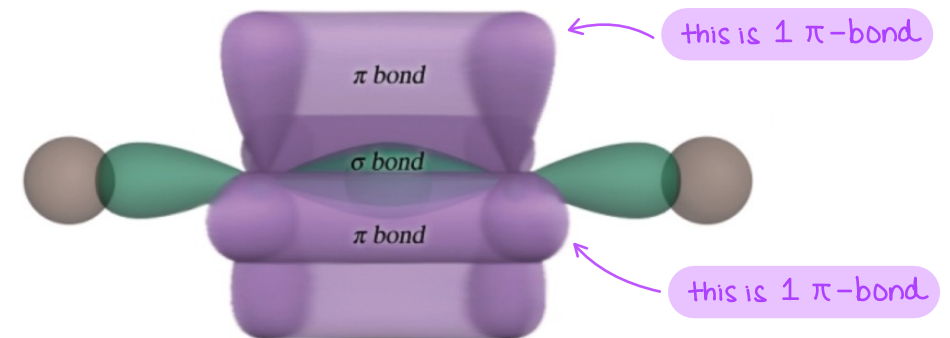


FIGURE 11.30 Two π -bonds in $\text{HC}\equiv\text{CH}$

IN-CLASS QUESTION 1

Learning Objective(s): 11.4, 11.5

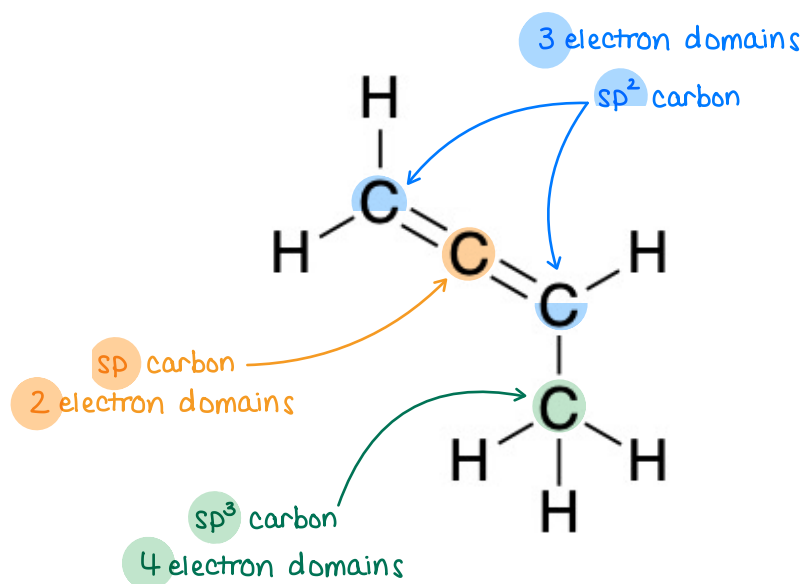
Which statements are *true* regarding hybrid orbitals?

- ☒ They have different energy values than the s - and p - orbitals from which they were derived.
- ☒ They have different shapes than the s - and p -orbitals from which they were derived.
- ☒ They help explain the shapes of molecules.
- ☒ They can be involved in the formation of σ -bonds.
- ☒ They can be involved in the formation of π -bonds. only through unhybridized p -orbitals!

EXAMPLE PROBLEM

Learning Objective(s): 11.4, 11.5

For the structure below, determine the total number of σ -bonds and π -bonds and identify the hybridization for each carbon atom.



Number of σ -bonds = 9

Number of π -bonds = 2

— single bond = 1 σ -bond

= double bond = 1 σ -bond + 1 π -bond

≡ triple bond = 1 σ -bond + 2 π -bond

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Identify the nonpolar molecules. Select all that apply.

Note: The Lewis structures are not intended to demonstrate molecular shapes. The elements A, E, and X may not represent the same elements in different molecules.

☐ Molecule 1

☐ Molecule 2

☐ Molecule 3

☐ Molecule 4

☐ Molecule 5

